

12.632

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Problem

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For given 3×3 matrix $\mathbf{A}$ eigenvalues are 1,2,3 then value of $\text{trace}(\mathbf{A}^{-1})$ is.

Let A be an invertible $n \times n$ matrix with eigenvalue λ and corresponding eigenvector $\mathbf{v}$, i.e.,

$$A\mathbf{v} = \lambda\mathbf{v}$$

where $\lambda \neq 0$ and $\mathbf{v} \neq \mathbf{0}$.

- The eigenvectors of A^{-1} are the same as those of A .
- The eigenvalues of A^{-1} are the reciprocals of the eigenvalues of A , i.e., $\frac{1}{\lambda}$.

Starting from the eigenvalue equation:

$$A\mathbf{v} = \lambda\mathbf{v}$$

Since A is invertible and $\lambda \neq 0$, multiply both sides by A^{-1} :

$$A^{-1}A\mathbf{v} = A^{-1}\lambda\mathbf{v}$$

$$\mathbf{v} = \lambda A^{-1}\mathbf{v}$$

Rearranging:

$$A^{-1}\mathbf{v} = \frac{1}{\lambda}\mathbf{v}$$

This shows that $\mathbf{v}$ is an eigenvector of A^{-1} with eigenvalue $\frac{1}{\lambda}$.

The eigenvalues of A^{-1} are the reciprocals of the eigenvalues of A , and both matrices share the same eigenvectors.

Let A be an $n \times n$ diagonalizable matrix. Then there exists an invertible matrix P and a diagonal matrix D such that

$$A = PDP^{-1}$$

where

$$D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$$

and $\lambda_1, \lambda_2, \dots, \lambda_n$ are the eigenvalues of A .

Taking the trace on both sides, we have

$$\text{tr}(A) = \text{tr}(PDP^{-1}).$$

Using the cyclic property of trace, which states that $\text{tr}(XYZ) = \text{tr}(ZXY) = \text{tr}(YZX)$ for any matrices X, Y, Z where the product is defined, we get

$$\text{tr}(PDP^{-1}) = \text{tr}(DP^{-1}P) = \text{tr}(DI) = \text{tr}(D).$$

Since D is diagonal, its trace is the sum of its diagonal entries, i.e., the sum of eigenvalues of A :

$$\text{tr}(D) = \lambda_1 + \lambda_2 + \cdots + \lambda_n.$$

Therefore,

$$\boxed{\text{tr}(A) = \sum_{i=1}^n \lambda_i.}$$

This completes the proof.

Let A, B, C be $n \times n$ matrices. The trace of the product ABC is defined as the sum of the diagonal elements:

$$\text{tr}(ABC) = \sum_{i=1}^n (ABC)_{ii}.$$

Recall that matrix multiplication components are given by

$$(ABC)_{ii} = \sum_{j=1}^n (AB)_{ij} C_{ji} = \sum_{j=1}^n \sum_{k=1}^n A_{ik} B_{kj} C_{ji}.$$

Thus,

$$\text{tr}(ABC) = \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n A_{ik} B_{kj} C_{ji}.$$

By changing the order of summation and renaming indices, we can write:

$$\text{tr}(BCA) = \sum_{j=1}^n \sum_{k=1}^n \sum_{i=1}^n B_{kj} C_{ji} A_{ik} = \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n A_{ik} B_{kj} C_{ji} = \text{tr}(ABC).$$

Similarly,

$$\text{tr}(CAB) = \sum_{k=1}^n \sum_{i=1}^n \sum_{j=1}^n C_{ji} A_{ik} B_{kj} = \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n A_{ik} B_{kj} C_{ji} = \text{tr}(ABC).$$

Hence, we have shown that

$$\text{tr}(ABC) = \text{tr}(BCA) = \text{tr}(CAB).$$

This proves the cyclic property of trace for three matrices. The property extends similarly to any product of matrices where multiplication is defined.

Solution

Now using properties mentioned above : eigen values of $\mathbf{A}^{-1}$ are $\frac{1}{1}, \frac{1}{2}, \frac{1}{3}$

$$\text{trace}(\mathbf{A}^{-1}) = \frac{1}{1} + \frac{1}{2} + \frac{1}{3} = \frac{11}{6} \quad (1)$$